

Note on gpt.py computational workflow.

- RMS norm: normalization of token's embedding vector $\mathbf{x} \in \mathbb{R}^d$. This is different from transformer paper's LayerNorm.

$$\text{RMS Norm}(\mathbf{x}) = \frac{\mathbf{x}}{\text{RMS}(\mathbf{x}) + \varepsilon}, \quad \text{RMS}(\mathbf{x}) = \sqrt{\sum_{i=1}^d x_i^2}, \quad \mathbf{x} = (x_1, \dots, x_d), \quad \varepsilon > 0.$$

- Rotary Positional Embedding (RoPE): positional embedding using sin, cos positional tensors. This is different from transformer paper's Positional Embedding.

Input x with shape (B, T, H, D) where B =batch size, T =sequence length, H =number of heads in multi-head attention, D =dimension of each head (must be even). Let $\sin = (\sin^{t,i})_{0 \leq t \leq T-1, 0 \leq i \leq D/2}$, $\cos = (\cos^{t,i})_{0 \leq t \leq T-1, 0 \leq i \leq D/2}$ be tensors each with shape $(1, T, 1, D/2)$. Then for each batch b , each index $t \in \{0, \dots, T-1\}$, each head h , RoPE does the following:

$$(x^{b,t,h,0}, \dots, x^{b,t,h,D-1}) \rightarrow \begin{cases} \cos^{t,i} x^{b,t,h,i} + \sin^{t,i} x^{b,t,h,i+D/2} & \text{for } 0 \leq i \leq D/2 - 1, \\ -\sin^{t,i-D/2} x^{b,t,h,i-D/2} + \cos^{t,i-D/2} x^{b,t,h,i} & \text{for } D/2 \leq i \leq D-1. \end{cases}$$

Position information, originally in t , is encoded in such rotation via cos, sin. Usually we take $\cos^{t,i} = \cos\left(\frac{t}{10000^{2i/D}}\right)$ and $\sin^{t,i} = \sin\left(\frac{t}{10000^{2i/D}}\right)$.

- Causal Self-Attention. The self-attention module with specific causal mask construction.

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